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The state of the art in key technologies for autonomous underwater vehicles: a review

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ABSTRACT

Autonomous underwater vehicles (AUVs) are becoming excellent tools for ocean exploration. Compared to other underwater vehicles, AUVs offer unique advantages in terms of autonomy, flexibility, and efficiency. They are used in a wide range of applications, including marine scientific research, resource exploration, and environmental monitoring. This paper reviews the current state-of-the-art developments in key AUV technologies, focusing on advancements in overall design, power systems, planning and control, perception, navigation, communications, artificial intelligence in AUVs, along with their applications. The various challenges encountered in advancing these key technologies are also summarized. This paper also examines future development trends in AUV technologies to enhance the operational efficiency and reliability of AUVs for increasingly complex tasks. By summarizing the recent research on key AUV technologies, this paper aims to strengthen research in these areas and contribute to fully leveraging the advantages of AUVs for marine system development and utilization.

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1. Introduction

The surface of the Earth is predominantly oceanic, with water bodies covering 71 % of the terrestrial surface. Oceans contain a wealth of resources, including minerals, chemicals, biological materials, and energy sources, which are economically and ecologically important. However, the current exploration and development of oceanic regions have only reached a mere 5 % of their total area, resulting in a significant number of unexplored or undeveloped oceanic regions [1,2]. To explore greater depths, a proportional increase in pressure tolerance is critical; consequently, deep-sea exploration by divers is currently untenable owing to safety considerations [3]. Practical methods for ocean exploration involve the use of underwater vehicles, which exhibit the potential to overcome the inherent limitations in the operation of aircraft and surface vessels. Although aircrafts can facilitate the rapid exploration of target areas, the detection signals they emit do not propagate effectively underwater. Surface vessels equipped with acoustic or

optical sensors can overcome certain issues related to signal transmission in water. However, they still face formidable challenges when conducting precise exploration in deep-sea regions. To address these challenges and achieve efficient and accurate ocean exploration, underwater vehicles have undergone significant developmental progress and are steadily becoming excellent tools for performing various activities at sea. These vehicles can reach specific depths and explore designated target areas, offering advantages in terms of detection capability and efficiency that aircrafts and surface vessels do not possess [4].

As underwater vehicle technology has progressed, various types of underwater vehicles have emerged. They can be categorized on the basis of whether they are manned or unmanned. Manned underwater vehicles include human-occupied vehicles (HOVs) [5]. Unmanned underwater vehicles (UUVs) include remotely operated vehicles (ROVs) [6], hybrid autonomous and remotely operated vehicles (ARVs) [7], autonomous underwater vehicles (AUVs) [8], and autonomous underwater gliders (AUGs) [9].

HOVs are human-operated underwater vehicles that reflect the comprehensive scientific capabilities of nations. With the increasing demand for marine scientific research, HOVs have become indispensable for tasks such as the study of deep-sea hydrothermal

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vents and mineral exploration because they enable proximate observations by marine scientists. The first HOV, Alvin, was developed by the Woods Hole Oceanographic Institution (WHOI) in 1964 [10]. In contrast, ROVs are unmanned and operated remotely from a support ship via umbilical cables. The Cable-Controlled Underwater Recovery Vehicle (CURV), which was the first ROV, was developed by the United States Naval Ordnance Test Station (now part of the Naval Air Warfare Center Weapons Division, NAWCWD) in 1960 and was used to search for lost hydrogen bombs on the seafloor. ARVs are a newer hybrid type of underwater vehicle that combines the characteristics of ROVs and AUVs, allowing both targeted operations and long-distance detection, thereby enhancing operational capabilities. The Nereus ARV developed by the WHOI is one of the most renowned ARVs, having dived into the Challenger Deep in 2009, thus becoming the first full-ocean-depth ARV [11]. AUVs can operate independently underwater without requiring umbilical cables and are known for their high degree of autonomy, wide operational range, and strong operational capabilities. The first AUV, Self-Propelled Underwater Research Vehicle (SPURV), was developed in 1957 at the University of Washington, USA [12]. AUGs are similar to AUVs but use buoyancy and attitude control for propulsion instead of conventional propellers or thrusters. They are highly efficient and consume minimal energy. The first AUG, ALBAC, was developed in 1992 at the University of Tokyo, Japan [13].

2. Review of typical underwater vehicles

With increasing demand for ocean exploration, underwater vehicles have undergone rapid development and can now perform various underwater tasks. Therefore, this review focuses on typical underwater vehicles based on their recent performances and describes the characteristics of each vehicle.

2.1. Human-occupied vehicles

Among the existing HOVs, Alvin is one of the most renowned. Since its commissioning, Alvin has undergone regular upgrades, with the latest having been completed in 2021. It is considered the most widely deployed and technologically advanced HOV worldwide. The Jiaolong HOV, developed by the China Ship Scientific Research Center, is the first manned submersible that was independently designed and built by China. On August 18, 2024, Jiaolong completed its 300th dive into the western Pacific Ocean [14]. The Russian manned submersibles Mir-1 and Mir-2, built in 1987, can dive to a maximum depth of 6000 m. In 2007, Mir-1 and Mir-2 conducted extreme dives under a 2 m thick layer of ice and completed a dive at the North Pole [15]. The Fendouzhe full-ocean-depth HOV, also developed by the China Ship Scientific Research Center, has an import substitution rate of 96.5 %. In 2020, Fendouzhe established a new Chinese record for manned deep-sea diving by successfully reaching the bottom of the Mariana Trench at a depth of 10 909 m. It currently holds the record for the most dives at a depth of 10 000 m [16]. The Shinkai 6500 HOV, developed and operated by the Japan Agency for Marine-Earth Science and Technology (JAMSTEC), was completed in 1989 and upgraded in 2012. It has been used in surveys of seafloor bathymetry, geology, and marine biology in the Pacific, Atlantic, and Indian Oceans [17].

2.2. Remotely operated vehicles

The Kaiko ROV, owned by JAMSTEC, successfully descended to 10 911 m in the Mariana Trench in 1995, setting a record for the deepest ROV dive at that time. However, in 2003, this ROV was lost because of the failure of two cables connecting it to its release

mechanism [18]. The Haima ROV, developed by the Shanghai Jiao Tong University, has an import substitution rate of 90 %. It can operate at a maximum depth of 3500 m [19]. The Haixing 6000, developed by the Shenyang Institute of Automation, Chinese Academy of Sciences, is the first ROV designed by China for depths of 6000 m. Through extensive scientific research operations, it has completed several missions at this depth, demonstrating the stability and reliability of this underwater vehicle [20]. The Jason ROV, operated by WHOI, can dive to a maximum depth of 6500 m. Its first dive was conducted in 1988. This ROV has performed hundreds of dives to survey hydrothermal vents. In 2016, the third generation of Jason became operational [21].

2.3. Autonomous and remotely operated vehicles

The Haidou-1 ARV, developed by the Shenyang Institute of Automation, Chinese Academy of Sciences, is the first full-ocean-depth underwater vehicle developed by China. It can withstand pressures at depths of up to 11 000 m and has completed dives exceeding 10 000 m. It collected the first sets of full-ocean-depth temperature and salinity data and real-time seafloor video data [22]. The Nereus ARV, developed by WHOI, can operate in both AUV and ROV modes. In 2009, it completed a dive to 10 902 m in the Mariana Trench, becoming the first UUV worldwide to reach such a depth [23]. The Rainbowfish ARV, which was developed by the Rainbowfish Ocean Technology Co. Ltd., was built with the help of social contributions. It can withstand pressures of up to 11 000 m [24]. The Ariane ARV, developed by the French Research Institute for Exploitation of the Sea (Institut Français de Recherche pour l'Exploitation de la Mer, IFREMER), can reach depths of 2500 m. It was commissioned for scientific cruises in early 2017, after six years of development and technical sea trials, and had completed 130 dives by the end of 2018 [25]. The Seaeye Sabertooth ARV, developed by Saab Seaeye, was designed for remote intervention, maintenance, and repair tasks. It can be recharged underwater via docking stations, significantly reducing or eliminating the need for support vessels [26].

2.4. Autonomous underwater vehicles

The full-ocean-depth Wukong AUV developed by Harbin Engineering University (China), set a world record for AUV deep diving in 2021 with a dive depth of 10 896 m in the Mariana Trench. It also achieved a linear distance of more than 15 km from the mothership for high-speed acoustic communication [27]. The Autosub Long Range 1500 (ALR1500), developed by the National Oceanography Centre, UK, is an ultralong-range AUV capable of extended underwater missions. In 2022, it demonstrated its underwater endurance capabilities by completing a continuous underwater journey of 2000 km [28]. The HUGIN Superior, developed by Kongsberg Maritime of Norway, is an AUV designed for geodetic, hydrographic, and marine archaeological missions. It is currently used by several navies worldwide [29]. Originally developed by WHOI, the Remote Environmental Measuring Units (REMUS) 6000 is an AUV that comprises various environmental sensors. In 2009, it was involved in the search for Air France Flight 447 and successfully located wreckages and crucial black boxes [30]. The Bluefin-21, developed by Bluefin Robotics (USA), is a highly modular AUV that can carry a variety of sensors and payloads. In 2014, it was used to assist in the search for Malaysia Airlines Flight MH370 [31].

2.5. Autonomous underwater gliders

The Petrel-L AUG, developed by Tianjin University, completed an underwater mission of 141 days and 3619.6 km in 2018, setting

new records for the longest continuous operation and distance covered by a Chinese AUG [32]. The Sea-Wing AUG, developed by the Shenyang Institute of Automation, Chinese Academy of Sciences, completed a synchronous observation mission with 12 AUGs in 2017, accomplishing the simultaneous operation of the largest number of similar Chinese AUGs [33]. The Slocum underwater glider, developed by WHOI, has been in operation since 1994. It has a maximum endurance of 40 000 km and uses thermoelectric energy for propulsion [34]. The Seaglider, the first generation of underwater gliders developed by the University of Washington, USA, is housed in a hydrodynamic fairing made of fiberglass. It is characterized by a hydrodynamic shape with low drag [35].

As mentioned previously, compared to other countries, the achievements of China involving HOVs, ARVs, and AUGs are significant, resulting in the development of multiple vehicles of these types with varying operational depths. Globally, China is a leading developer of submersibles that are capable of reaching depths of up to 10 000 m. However, although the Chinese ROV technology has developed rapidly, the ROV industry still has significant room for growth compared to the developments in other countries. Thus, developing the industrialization of ROVs in China is an ongoing requirement. In terms of operating depth, capability, and endurance, China has produced state-of-the-art AUVs. However, although Chinese AUV activities are primarily focused on scientific research on key technologies, other countries have already adopted AUVs as advanced tools for ocean exploration. Chinese AUVs still require considerable research and development before achieving widespread engineering applications.

2.6. Review of key technologies in AUVs

AUVs are essential tools in modern underwater exploration and research. AUVs are particularly noted for their capability to either execute tasks based on predefined plans or, in the case of more advanced platforms, to react to the environment and replan their missions in a fully autonomous mode, without the supervision of a human operator.

This independence significantly enhances the efficiency and safety of underwater operations, enabling AUVs to function in complex, hazardous, and hard-to-reach environments. Moreover, AUVs are outfitted with a variety of sensors to gather data on seawater temperature, salinity, depth, and pressure and to perform sonar imaging and bathymetric mapping. This versatility enhances the efficiency and quality of data collection, enabling AUVs to perform multiple tasks simultaneously.

AUV technology is a multidisciplinary field encompassing various areas. The key technological system components of an AUV are shown in Fig. 1. This review of key AUV technologies is organized as follows: Section 2 covers existing types of underwater vehicles and discusses key AUV technology systems; Section 3 provides the overall design of AUVs; Section 4 examines AUV power systems; Section 5 explores AUV planning and control technologies; Section 6 reviews AUV perception systems; Section 7 addresses navigation and positioning systems; Section 8 details AUV underwater communication systems; Section 9 introduces artificial intelligence (AI) in AUVs; Section 10 introduces the applications of AUVs; finally, Section 11 outlines the conclusions and future research on AUVs.

3. Overall system design

A factor considered in the overall design of an AUV is its shape. Decisions regarding shape design are based on the mission requirements and operational scenarios of the AUV to ensure effective

underwater operations. The choice of pressure hull depends on the operating depth and operational environment of the AUV, ensuring its safety during underwater operations. Descriptions of the common shapes and pressure hull designs in existing AUVs are provided subsequently.

3.1. Shapes

Most AUVs are torpedo-shaped. This design effectively reduces the drag during underwater operations and improves the endurance of the AUV. Examples include HUGIN 1000 AUV developed by Kongsberg Maritime [36], REMUS 100 AUV from WHOI [37], Bluefin SandShark AUV developed by Bluefin Robotics [38], and X300 AUV developed by the University of Pisa, Italy [39]. In addition to torpedo shapes, certain AUVs are designed with specific shapes for particular tasks, such as vertically flat and streamlined designs for rapid submersion. Examples include the full-ocean-depth Wukong AUV, Sentry AUV of WHOI [40], Odyssey IV class AUV of the Massachusetts Institute of Technology (MIT, USA) [41], and Xinghai 1000 AUV of Harbin Engineering University [42]. Biomimetic designs inspired by marine life have also been proposed, such as the manta-inspired AUV from Northwestern Polytechnical University (China) [43], Qianlong III AUV from the Shenyang Institute of Automation, Chinese Academy of Sciences, and U-CAT biomimetic AUV from Tallinn University of Technology (Estonia) [44]. Moreover, segmented AUVs have been designed to enhance stability during underwater operations. These include Autonomous Benthic Explorer (ABE) and SeaBED AUVs from WHOI [45,46], Girona 500 AUV developed by the University of Girona (Spain) [47], and Medusa deep sea AUV developed by the University of Lisbon (Portugal) [48]. As mentioned previously, the most common shape of AUVs is that of a torpedo, which facilitates better internal arrangement and deployment/recovery, thereby improving underwater operational efficiency. However, torpedo-shaped AUVs exhibit limited capabilities when AUVs are required to perform increasingly diverse underwater missions, such as deep diving, long-duration missions, and high-mobility expeditions. This has led to the development of AUVs with specialized shapes for specific tasks, including vertically flat, biomimetic, segmented, and framework designs. Therefore, future AUVs can be expected to follow two development paths: torpedo-shaped AUVs for general underwater tasks and specialized AUVs with unique shapes for specific missions. China has already developed AUVs in various shapes that can meet different underwater mission requirements. However, compared to other countries, research on AUV shape in China is relatively fragmented and lacks systematic planning. Therefore, the design process must be strengthened to enhance overall competitiveness.

3.2. Pressure hull

AUVs must withstand the pressure of water when submerged, which increases with depth. AUVs are designed with a pressure hull to protect the internal circuits and battery equipment. The pressure on the AUV increases with the increase in diving depth. Therefore, different pressure hulls are chosen for shallow and deep-sea environments to ensure safe underwater operation. In shallow areas, AUVs typically use cylindrical hulls to provide operational safety and improve hull efficiency. Examples include ALR1500, which has a maximum depth of 1500 m [49], AE2000a AUV developed by the University of Tokyo, with a maximum depth of 2000 m [50], and HUGIN 3000 AUV from Kongsberg Maritime, which can reach depths of up to 3000 m [51]. AUVs that are designed for deep-sea regions, where pressures are extreme, often use spherical pressure hulls. For instance, the full-ocean-depth Wukong AUV uses a spherical hull as its pressure hull. The

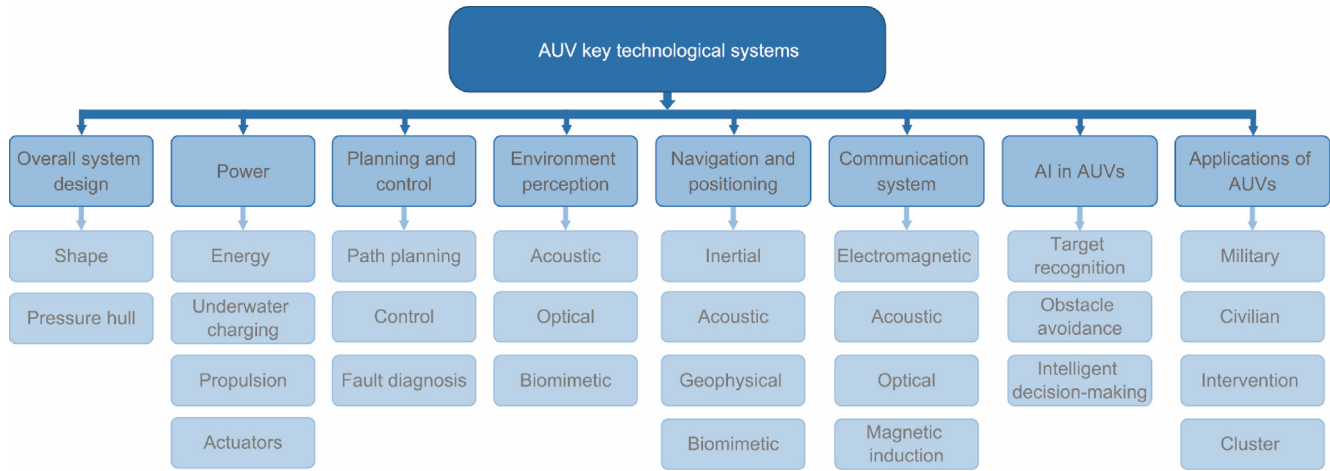


Fig. 1. Key technological system components of an AUV.

ALR6000 features a spherical pressure hull with a maximum depth of 6000 m [52]. The Dorado AUV developed by the Monterey Bay Aquarium Research Institute (USA) uses a combination of cylindrical and spherical pressure hulls [53].

In addition to the shape of the pressure hull, the material used for the hull is an important factor that is considered in the design. Even with the same shape, different materials can result in varying maximum diving capabilities for the pressure hull. Among the AUVs whose hulls comprise metallic materials, the Wukong 1000 AUV developed by Harbin Engineering University with a cylindrical titanium pressure hull has achieved a maximum diving depth of 1000 m. The Girona 500 AUV from the University of Girona has achieved a maximum diving depth of 500 m and uses a cylindrical 6082-T6 aluminum pressure hull. The Explorer AUV from International Submarine Engineering Ltd. (Canada) has achieved a maximum diving depth of 4000 m and uses a cylindrical pressure hull made of cast aluminum [54]. The ALR6000 AUV can dive to a maximum depth of 6000 m and uses a spherical aluminum pressure hull. Among the AUVs having hulls composed of non-metallic materials, the full-ocean-depth Wukong AUV has a maximum diving depth of 10 896 m and uses a spherical glass pressure hull. The Sentry AUV from WHOI has a maximum diving depth of 6000 m and uses a cylindrical alumina ceramic pressure hull. The Sea Whale 2000 AUV from the Shenyang Institute of Automation, Chinese Academy of Sciences, with a maximum diving depth of 2000 m, uses a cylindrical pressure hull comprising carbon fiber [55]. The Jellyfish Catcher AUV from the University of Tokyo has a maximum diving depth of 7300 m and uses a cylindrical silicon nitride ceramic pressure hull [56].

Currently, spherical pressure hulls provide optimal pressure resistance. Most deep-sea AUVs, such as the ALR6000, use spherical pressure hulls. However, owing to the uneven distribution of space within spherical pressure hulls, achieving an optimal utilization of the interior volume is challenging [57]. Cylindrical pressure hulls offer better space utilization but provide weaker pressure resistance [58]. Metals are the most commonly used materials for pressure hulls and offer advantages such as high strength, corrosion resistance, ease of processing, and high reliability. Most deep-water AUV designs rely on spherical metal pressure hulls to ensure adequate pressure resistance. Composite materials, such as ceramics and glass, offer higher pressure resistance, and cylindrical pressure hulls made from these materials can withstand immense pressure. Examples include the Sentry and Jellyfish Catcher AUVs. However, the processing and maintenance of composite materials is challenging. Therefore, future research should focus on advancing

composite material technologies, improving processing techniques, and ensuring that cylindrical composite pressure hulls can withstand the extreme pressures of deep-sea environments. Compared to other countries, China is at the forefront of research on the shape and materials of AUV pressure hulls, with a variety of hulls designed for shallow to deep oceans.

4. Power systems

The energy system stores the energy used to propel the AUV and power its equipment. The propulsion system generates the propulsive force required for the AUV motion, whereas the actuation system enables the turning, ascent, and descent phases of the AUV to perform underwater tasks. Next, we present the current research on energy, propulsion, actuation, and underwater energy-supply systems for AUVs.

4.1. Energy system

Batteries are the primary power sources for AUV motion within the energy system. Currently, Li-ion batteries are the most commonly used type of battery. With the advancements in Li-ion battery technology, an increasing number of AUVs are utilizing Li-ion batteries to sustain long-term underwater operations. Currently, two main types of Li-ion batteries are used: Li-ion and pressure-tolerant Li-polymer batteries. AUVs that use Li-ion batteries include REMUS 100, REMUS 600, Bluefin SandShark, and Teledyne Gavia, whereas AUVs that use pressure-tolerant Li-polymer batteries include HUGIN, Bluefin-9 M, Bluefin-12S, and Bluefin-21 [59]. In addition to the aforementioned Li-ion batteries, the gradual development of new energy vehicles has focused attention on LiFePO₄ batteries. Owing to their high working voltage, high energy density, long cycle life, and superior safety performance, LiFePO₄ batteries have garnered significant interest. Currently, several studies are focusing on LiFePO₄ batteries, primarily in the context of new energy vehicles. However, given the numerous advantages of LiFePO₄ batteries, their application in AUVs can significantly enhance their underwater operational efficiency. In their research on LiFePO₄ batteries for AUVs, researchers at the Shanghai Jiao Tong University examined the impact of deep-sea environments on the performance of power batteries in AUVs, and the effect of this impact on the accuracy of the remaining battery capacity estimation [60]. Researchers at the Southwest Jiaotong University proposed an intelligent battery management system for deep-sea LiFePO₄ batteries to enhance the long-term safety of deep-sea

energy storage devices [61]. Research on an emergency power system for AUVs was conducted at Temasek Polytechnic (Singapore) and a largescale 2 kW-h LiFePO₄ battery power system was proposed [62].

Given the limited energy density of Li-ion batteries and their lengthy charging times, high-energy-density fuel cells are being researched to enhance the underwater endurance of AUVs. JAM-STECH has been developing a fuel cell system for long-cruising AUVs. A 400 W high-efficiency multi-less fuel cell system successfully demonstrated 600 h of continuous power generation [63]. Polytechnic University of Turin (Politecnico di Torino, Italy) explored sedimentary microfuel cells for underwater power generation and suggested optimizing cylindrical sedimentary microfuel cells as a compact and cost-effective solution that is suitable for integration into AUVs [64]. ATLAS Elektronik GmbH (Germany) investigated a fuel cell system for the DeepC AUV using materials based on Nafion 112 for the membrane electrode assembly and gas diffusion systems [65]. Helion (formerly AREVA Renewables, now a subsidiary of Orano, France) also designed a fuel cell system for AUVs. Interest in this technology has increased after several successful experiments. For instance, URASHIMA AUV completed a 317 km continuous cruise [66]. AUVs require high endurance for underwater missions; however, the current energy density of Li-ion batteries cannot sustain long-term underwater operations. Fuel cells exhibit significant potential in offering greater endurance and more convenient energy replenishment. Therefore, future energy technologies for AUVs will likely focus on two primary development directions: increasing the energy density of Li-ion batteries and accelerating research on fuel cell technology. Compared to other countries, China is a leader in Li-ion battery research for AUVs and has developed LiFePO₄ batteries that are suitable for deep-sea environments. However, research in the field of fuel cells is limited, which must be strengthened to develop efficient underwater fuel cells.

4.2. Underwater charging system

Because AUVs rely on batteries for their energy supply during underwater operations, their limited energy often mandates frequent surfacing for recharging, which significantly reduces their operational efficiency. An underwater charging system can prevent AUVs from making frequent trips between their operating depths and the ocean surface, thereby effectively increasing their underwater endurance.

An underwater docking charging system for the REMUS 100 AUV was investigated by WHOI. Using an acoustic array and altitude information, the AUV follows a predefined path to an underwater wireless charging point [67]. Kawasaki Shipbuilding Corporation (Japan) proposed a coupling structure with a ferrite core and conical coils to improve the magnetic coupling device of the Marine Bird wireless charging system (Kawasaki Shipbuilding Corporation, Japan). The system can deliver 500 W of power with 96 % efficiency [68]. The University of Tokyo developed an underwater docking and charging system for AUV Tri-TON 2 (University of Tokyo, Japan). Harbor experiments were conducted under low visibility and strong currents, achieving a maximum underwater charging power of 400 W [69]. The Harbin Institute of Technology (China) proposed an approach that combines guidance and position locking to smoothly dock AUVs. A prototype has been built to validate the proposed method. It can deliver 1200 W to a load with 90 % efficiency [70]. Zhejiang University (China) provided a design process for selecting the optimal underwater operating frequency for inductive power transfer systems. The charging power of the AUV remained stable at 680 W with an efficiency of 90 % [71]. The Naval University of Engineering (China) proposed a novel open-type magnetic coupler with a simple structure, low weight,

and good self-inductance stability. When perfectly aligned, it can deliver 735.6 W with an efficiency of 90.87 % [72]. Underwater wireless charging technologies have become increasingly important as AUVs perform more underwater tasks. A key development focus in underwater wireless charging is achieving fast and efficient charging. Compared to other countries that have implemented wireless charging technology for AUVs, research in China is still in its theoretical and experimental stages. Further research is required before underwater wireless charging technology can be practically applied to AUV operations.

4.3. Propulsion system

Currently, propeller and water-jet propulsions are the main methods used in AUVs. Among these, propeller propulsion is the primary component of most AUV underwater propulsion systems owing to its simple structural design and control method.

ALR1500 is capable of executing long-duration underwater operations using an energy-efficient tail propeller, attaining a maximum speed of 1.2 m·s⁻¹ when the propeller power reaches 80 W [73]. The Chengsha AUV developed by Harbin Engineering University utilizes a tail thruster, two vertical thrusters, and two lateral thrusters for precise underwater control, with a maximum speed of 7 kn [74]. REMUS 100 also uses a tail propeller to drive the AUV with a maximum speed of 5 kn [75]. Gavia AUV is propelled by a tail propulsion system and has a maximum speed of 3 m·s⁻¹ [76]. The University of Tokyo developed a water-jet propulsion system for AUVs that focuses on creating a low-vibration monitoring platform with slow cruising speeds. This AUV can achieve a cruising speed of 1–2 kn [77]. Universiti Sains Malaysia developed a novel micro-AUV that uses a contractile water-jet thruster instead of a motor-driven rotary blade propeller to enhance stealth characteristics [78]. The Huazhong University of Science and Technology proposed a vector control method for water-jet propulsion in AUVs to address issues such as cavitation noise from propeller rotation and insufficient steering under certain conditions [79]. The Beijing Institute of Technology developed a hybrid thruster for spherical AUVs and conducted thrust measurements and basic motion experiments using underwater jets, propellers, and hybrid propulsion systems [80]. The advantages and disadvantages of the different propulsion methods are shown in Table 1.

As AUVs are required to perform increasingly complex tasks underwater, the demand for stability is increasing. The AUV propulsion system must not only enable fast movement but also ensure stable underwater operation. Moreover, because propulsion systems are a significant source of noise, achieving quiet underwater operations is a crucial area of research in the field of AUV propulsion systems. Overall, future AUVs will rely primarily on propeller propulsion to enhance their stability during underwater operations. Although the water-jet propulsion technology is structurally complex, it offers higher efficiency and lower environmental noise. It is suitable for specific underwater operations such as marine biological observations and the exploration of complex

Table 1
Advantages and disadvantages of propulsion methods.

System	Advantage	Disadvantage
Thruster	Low cost	Low efficiency
	High speed	Noise and vibration
	Simple structure	Poor maneuverability
	High adaptability	
Water jet	Efficient	Complex structure
	High speed	High maintenance costs
	Low noise and vibration	
	Good maneuverability	

underwater caves. In China, the research on AUV propulsion is not limited to thrusters; water-jet propulsion technology for AUVs has also been explored. However, research on water-jet propulsion is still in the theoretical stage, and considerable research is required before it can be applied to AUVs.

4.4. Actuator system

Stern planes, often referred to as rudder planes, are the primary actuators used to perform the turning and diving maneuvers of an AUV once it reaches sufficient speed relative to water owing to thruster activity. Based on their configuration, the stern planes can be categorized as either cross-shaped (cruciform) or X-shaped planes. In the cruciform design, the stern rudders are vertical and diving planes are horizontal. Rudder planes are often connected by a yoke, allowing them to turn simultaneously, which enables a single operating rod to control each pair of rudder planes. In contrast, the X-shaped rudder planes are more complex and use four rods that interact with each other. This design includes not only yaw control, but also four surfaces that simultaneously manage yaw and pitch. Each rudder plane has a separate control, and all four rods must move in unison to change the direction.

Maya AUV, developed by the Council of Scientific & Industrial Research- National Institute of Oceanography (CSIR-NIO), uses a cross-shaped configuration for control, achieving a travel distance of 21 km at $1 \text{ m}\cdot\text{s}^{-1}$ [81]. The Explorer 1000 AUV developed by the Shenyang Institute of Automation, Chinese Academy of Sciences also employs a cross-shaped configuration for control. Successful data acquisition using this vehicle has been instrumental in validating the three-dimensional (3D) spatial distribution characteristics of deep polar waters [82]. The SeaCat AUV, developed by Fraunhofer Institute of Optonics, System Technologies and Image Exploitation (Fraunhofer IOSB), uses a cross rudder to maneuver the AUV, enabling it to locate cracks or damaged areas in underwater structures or monitor critical subsea components [83]. The Explorer AUV is designed for underwater recharging and data transmission in ice holes. It utilizes an X-shaped configuration equipped with a remote return and short-range positioning system. The Thesus AUV, developed by International Submarine Engineering Ltd., also employs an X-shaped configuration and achieves a maximum diving depth of 1000 m. In 1996, the Thesus AUV successfully laid a 175 km fiber optic cable in the northern ice sheet of Ellesmere Island [84]. The Wukong 1000 AUV features an X-shaped configuration, a maximum diving depth of 1000 m, and a maximum speed of 6 kn. The HUGIN Superior AUV employs an X-shaped configuration and has achieved a maximum diving depth of 6000 m and can travel at 3 kn for three days. The advantages and disadvantages of the cruciform and X-shaped stern planes are shown in Table 2.

The cross- and X-shaped stern planes exhibit different performance characteristics. In a cross-shaped design, the rudders are often linked together by a yoke, exhibiting a simple design with good maneuverability. Compared to the cross-shaped arrangement, the X-shaped design is more complex and uses four rods that interact with each other. However, it provides better safety and

generates less noise than those in the former. As AUVs require increasingly stable control for the precise execution of underwater missions, advancements in actuator technology are likely to lead to more AUVs adopting X-shaped stern planes to provide stable operating conditions. The AUVs developed in China primarily use cross-shaped stern planes for AUV actuators, with only a few AUVs employing X-shaped stern planes. China must strengthen research on AUV actuators to further enhance their maneuverability.

5. Planning and control technologies

Planning and control technologies for AUVs include three main components: path or trajectory planning, control, and fault diagnosis. Path planning determines the path that an AUV should follow to avoid obstacles and reach a target point or to quickly and safely travel through a sequence of waypoints. In this case, the temporal constraints are not strict. Trajectory planning (also called trajectory generation) generates the (time-parameterized) trajectory that an AUV should track with the objective of meeting stringent temporal constraints. Control is essential for steering an AUV along a desired path or tracking a given trajectory in the presence of unknown external disturbances such as currents. Fault diagnosis identifies potential problems with sensors, actuators, and propulsion systems, and generates solutions to ensure smooth underwater operation that may include actuator reconfiguration for failsafe operations.

5.1. Planning

Motion planning for AUVs includes both path planning and trajectory generation. Path planning does not consider time factors and primarily focuses on determining a collision-free path from the start to endpoint. Conversely, trajectory generation incorporates time-dependent information into path planning, determining the speed and acceleration of the underwater vehicle during task execution to meet requirements such as smoothness and speed controllability.

5.1.1. Path planning

Path-planning methods for AUVs include global and local path planning. Global path planning is used when prior environmental information is available. Local path planning is used when the prior environment is either unknown or only partially known.

The A* algorithm is a static path search algorithm that combines heuristic and shortest path searches. A key component in guiding node selection is the specification of a cost function associated with the path [85]. Inspired by natural selection and evolution, genetic algorithms have been used for global path planning. This involves path encoding and fitness-function construction. The fitness function continuously updates the underlying population [86]. Differential evolution algorithms are used as path-planning optimization methods based on the evolutionary concepts of genetic algorithms. They are multi-objective optimization algorithms aimed at finding solutions in multidimensional spaces [87]. Particle swarm optimization algorithms achieve optimal solutions by continuously updating the velocity and position of an AUV until the stopping conditions are satisfied. These methods include particle swarms, fitness functions, and velocity updates [88]. Ant colony optimization (ACO) is a method for global path planning based on biomimicry. Inspired by the foraging behavior of ants, individual ant behavior is used to represent feasible solutions, and the collective behavior of the ant colony is used to form a solution space. ACO is based on evidence that ants exhibit intelligent behavior during foraging by sensing a chemical called pheromone [89].

Table 2
Advantages and disadvantages of stern plane configurations.

System	Advantage	Disadvantage
Cross rudder	Simple design	Larger surface
	Wide adaptability	Eddy current influence
	Good maneuverability	Large turning radius
X rudder	Good stability	Precise control
	Good lateral maneuverability	Complex structure
	Minimal eddy current effects	High power requirements

Rapidly exploring random tree is a random search method that constructs a random tree and randomly generates nodes in space. The algorithm starts from an initial point, creates random nodes, performs collision detection, and incrementally expands the tree until the distance from the generated node to the target endpoint is less than a specified threshold [90]. The artificial potential field method is a virtual force-based algorithm used to determine a suitable path. It creates an attractive force toward the target and a repulsive force away from obstacles within the motion space of the AUV. As the AUV moves, these forces vary, which helps it to avoid obstacles while approaching a target [91]. Fuzzy logic is a nonlinear, expert, and knowledge-based approach. AUV path planning involves fuzzification of sensor measurement data, fuzzy inference based on expert rules, and defuzzification to control AUV motion and achieve path planning [92]. Neural network methods, inspired by the human brain, are used to explore the patterns in data. For AUV path planning, a neural network model is constructed using sensor measurement data to train the network and output AUV motion information [93]. Reinforcement learning is an optimal learning strategy based on a trial-and-error approach based on environmental interaction for decision-making tasks. In AUV path planning, the AUV uses sensor data to detect the surrounding obstacles, estimate its current state, and change the environment. It receives reward signals from the environment to update its strategy [94].

5.1.2. Trajectory generation

The problem of trajectory generation is defined as the generation of a time-parametrized trajectory that a vehicle can track to meet spatial-temporal and/or energy-related objectives, considering the constraints imposed by the maneuverability characteristics of the vehicle while avoiding collisions with fixed or moving obstacles. A wealth of literature is available on the problem of trajectory generation for autonomous vehicles, which has significantly benefited from seemingly disparate yet closely related contributions to the fields of aerial, land, and marine robotics, including AUVs [95–98]. Several methods described in Refs. [95–98] formulate the problem of trajectory generation using the classical tools of optimal control theory. In this setup, given a model of an autonomous

vehicle, the problem is converted into an optimization problem by proper choice of the vehicle inputs and an energy- or time-related objective function, subject to state, actuation, and collision-avoidance constraints. The optimal control problems arising in this context are generally difficult or impossible to solve analytically, and numerical methods for computing adequate solutions must be sought. Among the latter, indirect methods apply the necessary conditions for the optimality in an infinite problem to derive a boundary value problem in ordinary differential equations, whose solutions can then be determined numerically [95]. In striking contrast to indirect methods, direct methods are based on transcribing infinite optimal control problems into finite-dimensional nonlinear programming problems (NLPs) using a discretization scheme [99]. The latter problem can be solved using readily available NLP solvers. Over the years, several different discretization schemes have been used, including pseudo-spectral methods [100] and Bernstein polynomial approximants [96,97]. The latter have only recently come to the fore; however, owing to their powerful properties, they hold considerable promise for the close to real-time implementation of trajectory generation systems for single and even multiple autonomous vehicles working in cooperation [96]. The advantages and disadvantages of the various planning methods are listed in Table 3.

The challenges in AUV planning algorithms are growing as AUVs encounter more complex underwater tasks. To obtain better routes, these algorithms must adapt to increasingly complex scenarios and consider increasing number of factors during planning. In current planning methods, AI technology can overcome the limitations of system dimensions and determine an optimal path while considering multiple factors. With the gradual improvement in the performance of computers installed on AUVs, the use of AI technology for online planning in AUVs is demonstrating considerable potential.

5.2. Control methods

The motion control of AUVs encompasses a wide spectrum of problems ranging from simple vehicle primitives such as pose (position and attitude), depth, and altitude control to more advanced

Table 3
Advantages and disadvantages of path-planning methods.

Type	Method	Advantage	Disadvantage
Global path planning	A*	Direct search Less time required Low computational complexity	Not the optimal solution Grid resolution affects computational complexity
	Genetic algorithm	Global search	Lack of stability Trapped in local optima
	Differential evolution algorithm	Global search	More likely to mutate
	Particle swarm optimization algorithm	Quick search	Local optima trapped
	ACO	Efficient Good robustness	Local optima trapped
Local path planning	Rapidly exploring random tree	Easy pathfinding High environmental adaptability	Low search efficiency Poor path-optimizing ability
	Artificial potential field	Simple structure Good real-time performance	Unstable path Local optima trapped
	Fuzzy logic	Good applicability Low calculation effort	Reliance on experts Lack of system robustness
	Neural network	Simple learning rules Strong learning ability	Long training time Difficult to obtain samples
	Reinforcement learning	High robustness High generalization ability	Dimensionality issues Handcrafted feature conditions
Trajectory generation	Direct methods	Handling nonlinear constraints Suitable for high-dimensional problems	Local optima trapped Challenges in balancing accuracy and efficiency
	Indirect methods	High precision High adaptability	High computational load

primitives that include waypoint, path following, and trajectory tracking control. Trajectory tracking refers to the problem of making an AUV track a time-parameterized curve. Path following does not involve explicit temporal constraints and aims to steer an AUV toward a desired path and travel along it with a speed profile that may vary along the path [101,102]. That is, in contrast to path following, an AUV must be at specific positions at designated times in the trajectory tracking process. In essence, the design of any control primitive can primarily be reduced to steering an appropriately defined generalized error to zero. Numerous methods have been proposed for AUV motion control, and the final choice of method depends on the complexity of the problem (e.g., linear versus nonlinear, robust versus adaptive).

Currently, the primary methods used to control AUVs include proportional–integral–derivative (PID) control [103], sliding mode control (SMC), adaptive control, model predictive control (MPC), and neural network control. PID control is one of the earliest control strategies and is known for its simplicity and reliability. It constructs an error (deviation of the actual output of a plant from a desired set point) and issues commands to the actuators to reduce the error to zero. This error is processed using proportional, integral, and derivative terms. However, changes in the operating environment and payload variations can affect the control performance, impeding PID control from satisfying the accuracy requirements [104]. Conceptually, SMC is a nonlinear control characterized by the fact that it attracts the state of a system to a properly designed surface, after which it forces it to move along this (sliding) surface toward the origin. SMC exhibits several advantages, including fast response and increased robustness with respect to parameter variations. However, once the state trajectory reaches the sliding surface, it may oscillate around it, leading to chattering problems [105]. In practice, this problem is alleviated by introducing a boundary layer to smooth the activity of control commands.

Adaptive control can be categorized as model-based, indirect, or self-tuning control. Adaptive controllers adjust their parameters in real time based on perceived changes in the external environment or internal vehicle parameters, leading to improved control performance. Adaptive control requires knowledge of the mathematical model of the target, which makes the design more complex [106]. MPC is based on the dynamic model of the process, optimizing a given cost functional over a sliding finite-time window and applying, at each time step, the first component of the resulting optimal input history, after which the process repeats itself. It exhibits the advantages of ease of modeling and strong robustness; however, when system complexity is high, it may encounter issues with decreased real-time performance [107]. Neural network control is a branch of intelligent control that uses neural networks as either identifiers within the control structure or primary controllers for systems with uncertain mathematical models. Currently, neural networks are computationally complex and require high hardware performance for AUVs [108]. The advantages and disadvantages of these control methods are listed in Table 4.

As the range of underwater missions of AUVs increases, the demand for advanced control technologies also increases. Although traditional control methods can manage AUVs, they do not satisfy the requirements of modern AUVs. AI technology shows promising potential for achieving robust AUV control. Although the direct application of AI to AUV control presents several challenges, its future development potential is considerable.

5.3. Fault diagnosis

The purpose of fault diagnosis is to determine whether a motion actuator or sensor has failed, identify the specific location of the fault, and assess the type and severity of the fault to improve the

Table 4

Advantages and disadvantages of control methods.

Method	Advantage	Disadvantage
PID	Simple structure Easy to implement and adjust	Insufficient for nonlinear systems
SMC	Quick response Increased robustness to parameter variations and disturbances	May cause chattering
Adaptive control	Strong robustness Superior control effect	Convergence issues Long parameter estimation time
MPC	Simple modeling Strong robustness	Real-time performance decreases with system complexity
Neural network	Strong robustness Effective in handling model uncertainty	Computational complexity Insufficient convergence in online learning

safety of underwater AUV operations. In the specific case of partial actuator failures, fault diagnosis may facilitate actuator reconfiguration to enable failsafe operations. Currently, the primary methods used for fault diagnosis include signal processing and model analysis, as well as techniques that are based on prior knowledge or data statistics.

Signal processing methods extract features such as amplitude, frequency, and phase from directly measurable input and output information and analyze their trends for fault diagnosis. However, such methods have limited applicability because the measurement signals from AUVs are often noisy owing to environmental disturbances [109]. The model analysis method involves modeling systems that may undergo failures and performing fault diagnosis by comparing their (system state) estimates with the actual measured conditions. However, AUVs are strongly coupled with multiple degrees of freedom, complicating the process of obtaining accurate models [110]. The prior-knowledge-based method involves qualitative analysis based on the operating mode, failure properties, and responses of the system being diagnosed [111]. The data-statistics-based method uses relevant data from the fault and normal states of the diagnosed system to train fault diagnosis models using machine learning. Machine learning methods exhibit significant potential in fault diagnosis. However, the challenge with this method is that faults are rare events, making it difficult to obtain sufficient fault samples for model training [112]. The advantages and disadvantages of fault diagnosis methods are listed in Table 5.

Fault diagnostic technology is crucial for achieving accurate control because AUVs require precise underwater control. Moreover, obtaining more fault information through diagnostic technology provides valuable reference information for subsequent

Table 5

The advantages and disadvantages of fault diagnosis methods.

Method	Advantage	Disadvantage
Signal processing	Efficient Easy to implement	Lack of stability Easily affected by noise
Model analysis	Easy to analyze	Difficult to achieve accurate modeling
Prior knowledge	Efficiency Accuracy	Limited scope Significant subjective factors
Data statistics	High precision Handling of nonlinear systems	Difficult to obtain large amounts of data

control tasks, ensuring that the precise control of the AUV can be maintained, even in the event of a fault. Among the current fault diagnosis methods, data-statistics-based methods that utilize AI technology can effectively identify both the type and severity of faults. As AUV technology advances and more data become available for training, the accuracy of fault diagnosis is expected to improve significantly, demonstrating promising application prospects.

6. Environment perception system

The perception system of an AUV serves as its “eyes,” and is responsible for sensing the environment, ensuring safety, and performing sensor-based underwater tasks. It can be categorized into acoustic, optical, and biomimetic perceptions based on the different principles of underwater sensing.

6.1. Acoustic perception

Acoustic signals can effectively transmit data underwater, and various acoustic sensing devices are used, including multibeam echo sonars (MBESs), single-beam echo sonars (SBESs), side-scan sonars (SSSs), and synthetic-aperture sonars (SASs). Among these, the MBES is the most commonly used acoustic sensor for underwater mapping. They emit acoustic waves in a fan shape underneath a transceiver and use beamforming to extract directional information from the reflected sound wave. They can provide high-accuracy depth data with a significant data volume and resolution. For instance, the JINBEI (developed by JAMSTEC) [113] and Dorado AUVs utilize MBESs for seafloor bathymetric measurements [114]. The Norwegian Defence Research Establishment (FFI) conducted shallow-water AUV terrain-aided navigation (TAN) trials using an MBES as the primary environmental perception sensor [115]. The SBES operates on the same principle as the MBES, sending acoustic beams to capture seafloor bathymetry. At the same operating frequency, the SBES uses significantly less power than the MBES, effectively extending the underwater endurance of the AUV. Both the ALR1500 and Chengsha AUVs employ SBES for TAN [116]. The University of Florence (Italy) used bottom height data obtained from a Doppler velocity log (DVL) for bathymetric mapping of the seafloor. Moreover, the feasibility of the method that was used in the FeelHippo AUV (developed by University of Florence) equipped with a DVL was verified through sea trials [117]. The SSS is an underwater imaging sonar that differs from the MBES and SBES in its measurement approach. The SSS generates scanned line intensity information of the seafloor terrain. The REMUS 100 and Bluefin SandShark AUVs use SSS to measure the seafloor terrain profile. The University of Florence utilized the FeelHippo AUV equipped with a forward-looking sonar (FLS) to identify various features of underwater areas and roughly estimated the underwater terrain using the FLS and altimeter [118]. Geometric correction methods for the SSS images, including range and velocity corrections, were investigated in Ref. [119]. The high-resolution underwater imaging technique known as SAS is based on a small-aperture array and the movement of this array to create an equivalently large aperture. The FFI used the high-resolution interferometric synthetic aperture sonar (HISAS) 1030 SAS for TAN. At the beginning of the mission, when the navigation accuracy was high, the AUV performed autonomous mapping by measuring baselines and creating baseline maps [120].

6.2. Optical perception

Optical perception devices, particularly underwater cameras, are another method of perception in addition to acoustic devices.

Currently, the primary types of underwater cameras are monocular, stereo, and fisheye cameras.

The monocular camera has only one lens. It processes and calculates images based on the data captured by the lens, thereby enabling functions such as scene acquisition and tracking. The University of Florence has developed and evaluated a navigation method for AUVs using a monocular camera. These algorithms were tested using image sets obtained from the Zeno AUV (University of Florence, Italy) in Haifa Bay, Israel and FeelHippo AUV in Vulcano, Italy [121]. Researchers at the University of Florence proposed a monocular visual odometry algorithm designed for underwater robot navigation. The algorithm was tested using a dataset collected by the compact Zeno AUV in Haifa Bay, Israel [122]. A stereo camera is a machine-vision technology that captures two images of an object from two different positions. A vision-based navigation strategy for AUVs was proposed by Harbin Engineering University. A series of virtual and real experiments demonstrated the reliable docking navigation of the Yuxiang AUV (Harbin Engineering University) [123]. The Dalian University of Technology proposed a robust real-time self-localization method for AUVs based on stereo cameras and inertial sensors. The experimental results showed the effectiveness of the proposed method in dynamically changing underwater scenarios [124]. Fisheye cameras are specialized cameras with a significantly short focal length and field of view of approximately 180°. Shanghai University proposed an imaging system for binocular vision based on fisheye lenses. The experimental results demonstrated that the corrected underwater fisheye images had a lower average reprojection error [125]. The University of Pisa proposed an underwater navigation framework based on the use of stereo cameras as linear velocity sensors. The data collected by the SPARUS II AUV (University of Girona) were used for verification, and the results showed that, compared to the DVL benchmark, the obtained results confirmed the feasibility of this method for underwater inspection [126]. The University of Michigan (USA) proposed a new vision-based underwater scene mapping method for robotic operating systems. The results from these datasets demonstrated the high accuracy of the system and its applicability in diverse natural environments [127].

6.3. Biomimetic perception

Underwater biomimetic perception technology refers to techniques that enhance the underwater detection and perception capabilities by simulating or emulating natural sensory systems. This technology has a wide range of potential applications and can facilitate more efficient detection, identification, and navigation in underwater environments. Currently, artificial lateral line system (ALLS) and biomimetic electric field perceptions are common underwater biomimetic perception technologies.

The lateral line is a specialized sensory organ in fish that helps them perceive information about their underwater environment. Inspired by this, the design and development of ALLS for AUVs have become prominent topics. The Ocean University of China developed a practical AUV with an ALLS. A neural network model was used to identify the coordinates, frequency, and amplitude of a vibration source, achieving an identification accuracy of 93 % [128]. Harbin Engineering University combined data from an inertial measurement unit (IMU) and the ALLS to propose a fusion method based on factor graph optimization. In tank experiments, this method demonstrated a significant improvement in the trajectory estimation capability of an AUV [129]. Using ALLS probes for aquatic habitat assessment, researchers at the University of Valladolid (Spain) effectively captured complex information, reduced sampling effort, and outperformed traditional methods, indicating a more efficient and accurate approach to fish habitat modeling

[130]. Inspired by the perception behavior of weakly electric fish, researchers have studied the principles and methods of weak electric field-based underwater perception. An underwater motion target localization method based on active electric field sensing principles and long short-term memory networks was proposed by Xi'an Jiaotong University. The two-dimensional (2D) mean absolute and relative position errors were 5.38 mm and 1.06 %, respectively [131]. To enhance the performance of underwater electric field detection systems, Harbin Engineering University proposed an underwater detection method based on rotating electric field theory. The development of a prototype and experiments to detect metal objects in aquatic environments have been completed, laying a foundation for further research [132]. The advantages and disadvantages of the various perception systems are shown in Table 6.

Underwater sensor technology is advancing along with AUV technology. Long-range, accurate, and quiet detection technologies continue to be the primary focus of underwater sensing research, aimed at ensuring the stealthy operation of AUVs and minimizing their impact on surrounding marine life. Compared to other countries, China entered the field of underwater perception at a later stage. Although many advanced underwater sensors have been developed, most acoustic or visual perception sensors used in Chinese AUVs, particularly acoustic sensors, originate in other countries. A gap remains between China and other countries in terms of the development of underwater perception sensors, and improving the stability of these sensors is necessary to enhance international competitiveness.

7. Navigation and positioning system

Underwater operations require AUVs to operate beneath the surface. The global navigation satellite system (GNSS) cannot provide real-time position information for AUVs underwater because electromagnetic waves rapidly weaken in such environments. Therefore, with the unavailability of the GNSS, AUVs have to rely on alternative underwater navigation methods. The current methods for AUV navigation include inertial, acoustic, geomagnetic field, and biomimetic navigation methods.

7.1. Inertial navigation system

The most commonly used underwater navigation method for AUVs is the inertial navigation system (INS) [133]. It employs onboard inertial sensors such as gyroscopes and accelerometers to calculate the underwater position of the AUV. The accelerations and angular accelerations provided by these sensors are used to

estimate the subsequent position of the AUV based on the initial position information [134], depending on how the acceleration data are processed. To determine the position of the AUV, the INS integrates measurements from gyroscopes and accelerometers. However, owing to measurement errors in these sensors, the INS accumulates errors over time, resulting in imprecise underwater navigation over long distances without external aids. Therefore, it is typically used as the primary navigation device. Northwestern Polytechnical University proposed an improved factor graph and used the Mahalanobis distance for anomaly detection to address the challenges of underwater vehicle navigation by combining INSs, DVL, and ultrashort baseline (USBL) [135]. Northwestern Polytechnical University also proposed an alternative state-independent underwater fusion invariant filtering method using DVL, IMU, and a two-frame configuration in the Earth-centered Earth-fixed coordinate system [136].

China is currently capable of independently developing and designing INSs. However, domestic INSs still lag behind international leaders in terms of their accuracy and stability. As domestic sensor technology continues to advance, the capabilities of INSs developed in China are expected to align more closely with those of international leaders.

7.2. Acoustic navigation

Acoustic signals are less affected by seawater, and can transmit information over long distances. Acoustic navigation involves the use of multiple underwater acoustic arrays with communication between the arrays referred to as baselines. Based on geometric principles, underwater positioning methods measure the distances and/or bearing angles between an AUV and acoustic array to determine the position of the AUV. Acoustic navigation systems can be categorized into long baseline (LBL), short baseline (SBL), and USBL systems based on the length of the baseline [137]. LBL navigation requires preinstalled acoustic arrays with known positions on the seafloor. The AUV transceivers exchange information with the array responders to determine the distance between them, enabling underwater positioning via trilateration. Although they are complex and expensive, LBL systems are often deployed in areas with flat seafloors and frequent ship and AUV traffic. In SBL navigation, multiple receivers and acoustic beacons on an AUV are used to intersect distances and determine positions. However, this method can be affected by the AUV deformation. USBL systems use multiple acoustic arrays and beacons to measure distance and direction. Although compact and convenient, their accuracy is dependent on the distance and less suitable for deep-sea navigation. The SBL and USBL systems are significantly affected by water depth, with the positioning accuracy decreasing as the depth increases. These systems require the precise calibration of transducers.

China has conducted extensive research on underwater acoustic navigation, and now possesses acoustic navigation technology that covers all depth ranges. However, compared to other countries, despite the development of key technologies on a global level, the degree of commercialization remains relatively low. Other countries have commercialized a wide range of underwater acoustic navigation products. Therefore, China needs to enhance its fundamental technological research and improve the stability of its acoustic navigation products.

7.3. Geophysical navigation

Underwater geophysical navigation uses spatially distributed geophysical fields to determine the position of an AUV. For this purpose, the geophysical information obtained from the AUV sensors are compared with an a priori database. Geophysical naviga-

Table 6
Advantages and disadvantages of perception systems.

System	Advantage	Disadvantage
Acoustic	Large-area detection Widely used Less affected by environment	Expensive Low resolution Poor performance in shallow water
Optical	Low cost High resolution Rich in detection information	Limited detection range Sensitive to light and water quality
Bionic artificial lateral line	High coverage Good environmental compatibility	Limited detection area Application needs improvement
Bionic electric	Capable of 3D detection Good environmental adaptability	Small detection area Limited detection range Needs improvement in application

tion can be classified into three types based on the information used: terrain-, magnetic-, and gravity-based methods. This type of navigation typically complements the INS by using geophysical data to periodically correct the drift of the INS, thereby enhancing long-range, high-precision underwater navigation.

TAN involves creating an a priori terrain map of the operational area using premeasured terrain data. This map is used to determine the position of the AUV and correct the INS drift by matching real-time seafloor terrain data with the a priori terrain map. Researchers at the Shanghai Jiao Tong University developed TAN methods for long-range AUVs based on cubature particle filter (PF). Simulations performed on actual bathymetric maps showed that the proposed method can improve the average positioning accuracy [138]. Gravity-aided navigation involves matching gravity anomalies or gravity gradient values with a premeasured gravity field map to determine the position of the AUV. It uses gravity-sensitive sensors to measure gravity and correct navigation drift. The Beijing Institute of Technology has proposed an enhanced gravity vector matching algorithm based on the PF. Simulation results indicate that this method provides high accuracy and robustness [139]. Magnetic-aided navigation is similar to TAN, but relies on matching magnetic sequences obtained through measurements with premeasured underwater magnetic maps to determine the position of the AUV. Unlike terrain information, magnetic navigation incorporates features such as magnetic declination, magnetic inclination, and up to seven different components to select the best matching features for underwater navigation. University of Lisbon proposed a seafloor magnetic navigation system that combines sequence estimation algorithms with complementary profile correlation methods. The Medusa autonomous surface vehicle was used for testing. The test results indicate that this navigation method can converge even with large initial errors, with a mean position error ranging from 10 to 20 m [140,141].

AUVs require a priori geophysical field data for underwater navigation using the aforementioned geophysical field navigation methods. However, these methods cannot be implemented in areas without a priori data. To address this limitation, simultaneous localization and mapping (SLAM) technology was introduced to eliminate the need for a priori geophysical field data. The University of Girona employed a newly developed AUV-mounted 3D laser scanner for real-time SLAM. A laser-scanning Girona 500 AUV (University of Girona) was used for the demonstration, and the results indicated the possibility of limiting the navigational drift and producing consistent high-precision 3D maps of the inspected targets [142]. The University of Florence proposed a SLAM algorithm framework that integrates a monocular camera and DVL information. Optical sensors are used for pose and landmark constraints, whereas DVL resolves unknown scale factors. The algorithm was evaluated using sea trials of the FeelHippo AUV [143]. The University of Haifa (Israel) proposed a SLAM method that combines sonar and optical images to achieve underwater navigation. This method matches objects detected by the camera with those detected in SAS images. Finally, the effectiveness of the algorithm was verified through simulation experiments [144]. AUVs typically follow predefined trajectories to perform underwater tasks using the aforementioned SLAM methods. However, if an AUV continuously passes through unsuitable terrain when using the SLAM technology, correcting its positioning results becomes difficult. Active SLAM technology has been proposed to address this issue. This technology aims to guide the AUV back to suitable areas, thereby improving the robustness of navigation results. Researchers at the Harbin Engineering University proposed an active bathymetric SLAM method, in which an AUV is actively guided into areas with significant terrain variations for

accurate loop detection. This method was shown to be more effective than the existing algorithms, as demonstrated by simulation experiments [145].

With the advancements in geophysical field measurement technology, research on geophysical field navigation has progressed rapidly, offering viable methods for precise long-term underwater navigation. Compared to geophysical field navigation technologies of other countries, China has achieved significant breakthroughs in key areas. However, while other countries have already widely implemented these technologies for AUVs, research in China is still in the experimental stage and requires further exploration before widespread application can be achieved.

7.4. Biomimetic navigation

Biomimetic algorithms are optimized search methods that mimic the growth patterns of natural species or replicate the activities of certain organisms. Magnetic biomimetic navigation is inspired by the use of magnetic fields by biological species for navigation. Extensive research has shown that some animals can use geomagnetic information for navigation. For example, sea turtles navigate using magnetic cues. Northwestern Polytechnical University proposed a biomimetic geomagnetic navigation method for AUVs that did not rely on prior geomagnetic information. This was achieved by constructing cost functions and considering the magnetic slip behavior of organisms [146]. The Beijing Institute of Technology proposed an improved elitist multi-objective artificial bee colony algorithm that incorporates information from magnetic anomalies. Simulations confirmed the effectiveness of this algorithm [147]. The National Institute of Ocean Technology has proposed a bio-inspired machine-learning-based geomagnetic field navigation method, which has the potential to replace GNSS for navigation based on a 50 m resolution geomagnetic map [148].

One type of biomimetic navigation is polarization light navigation, which is based on the underwater homing behavior of marine organisms, such as mantis shrimp. This method provides navigational information characterized by high real-time performance with no drift using a polarization model generated from sunlight entering the atmosphere and seawater interface. The University of Illinois Urbana-Champaign investigated precise underwater polarization navigation using biologically inspired polarization-sensitive imaging sensors to determine the geographic position of an observer based on radial underwater polarization patterns. Experimental data recorded at different depths and times around the world revealed an error of 6 m per kilometer traveled [149]. The Dalian University of Technology created simulation models of underwater polarized light patterns under various conditions and designed experiments to study the effects of different factors on these patterns. The feasibility of underwater polarized light navigation under various conditions was demonstrated by the consistency between the simulations and measurements [150]. The advantages and disadvantages of the navigation methods are listed in Table 7.

To accurately perform underwater missions, precise long-term underwater navigation technology for AUVs is crucial. However, no effective method is currently available for long-term precise underwater navigation of AUVs. Therefore, the development of low-energy methods for achieving precise long-term navigation is an important research focus in the field of AUV navigation.

8. Communication

During operations, AUVs must be able to transmit information underwater, either to other AUVs or surface vessels. Currently, the primary methods of underwater communication include elec-

Table 7
Advantages and disadvantages of navigation methods.

Method	Advantage	Disadvantage
Inertial	Good stability Technologically mature Wide range of applications	Accuracy is related to cost Errors accumulate over time
Acoustic	High navigation accuracy	Limited operational range High deployment and recovery costs
Geophysical	No cumulative errors No external sensors required	Technology not yet highly mature Navigation results affected by prior maps
SLAM	No cumulative errors No prior maps required	Limited positioning range Affected by operational area characteristics
Active SLAM	No cumulative errors Actively constrains errors	Technology is not yet mature Active backtracking is time consuming
Biomimetic	No cumulative errors Global navigation capability	Poor accuracy Insufficient robustness
Polarization light	No cumulative errors Strong real-time capability High navigation accuracy	Restricted by geophysical fields Affected by weather Complex technology

tromagnetic wave, acoustic, optical, and magnetic induction communication methods.

8.1. Electromagnetic communication

Although underwater optical communication offers a high data-transmission rate, light waves exhibit difficulty in passing through the air–water interface in cross-media communication scenarios, often requiring relays to forward the signal. In comparison, underwater electromagnetic wave communication offers distinct advantages. Electromagnetic waves can be transmitted directly from the base station to the underwater target and can pass through the air–water interface, significantly expanding the application range and supporting the establishment of a cross-media network system. Despite the rapid attenuation of electromagnetic wave signals underwater, they offer shorter delay times, faster data-transmission speeds, and widely accepted standards compared to other methods [151].

8.2. Acoustic communication

Underwater acoustic communication is the primary method for AUVs to communicate with the outside world because it is the only reliable method for long-range wireless information transmission underwater, covering distances of hundreds to thousands of meters. Tianjin University proposed a multi-AUV communication scheme based on an orthogonal frequency-division multiple-access downlink communication. The algorithm outperformed existing algorithms in terms of bit-error-rate performance based on simulation and sea-trial results [152]. Jiangsu University of Science and Technology (China) proposed an AUV-based underwater acoustic covert communication (UACC) game model to study the adversarial interactions between AUV-based covert transmitters and detectors in the Internet of Underwater Things. Simulation results show that the proposed solutions improve UACC performance compared to existing solutions [153]. Researchers at the Hebei University of Engineering (China) examined the offensive and defensive interactions between AUV jammers and a defender sensor network. Numerical results show that the proposed multi-AUV security transmission scheme based on a constrained budget

game outperforms the expected and utility benchmarks [154]. École de Technologie Supérieure (Canada) incorporated intelligent reflecting surfaces between the recruited nodes and AUVs to achieve more reliable underwater communication [155].

8.3. Optical communication

Underwater wireless optical communication (UWOC) technology, which is known for its high bandwidth potential, has become a focal point of research. However, establishing reliable UWOC links presents significant technical challenges owing to the complexity of the marine environment. Researchers at the King Abdullah University of Science and Technology (Saudi Arabia) investigated the tracking problem between AUVs and moving surface vessels, both equipped with optical communication transceivers. A channel model was approximated based on a conical light beam area to ensure a minimum bit rate if the AUV transmitter remained within this area [156]. McMaster University (Canada) proposed a new method for establishing a reliable high-speed broadcast communication link between a group of AUVs in sea ice. The results demonstrated that the system achieved a 100 Mbit·s⁻¹ transmission rate and 100 mW average power at distances of 3.5 and 3 m in 36 and 12 cm thick snow-covered and bare sea ice sheets, respectively [157]. The University of Porto (Portugal) introduced a multilens receiver scheme to enhance the angular misalignment tolerance between AUVs and sensor planes in UWOC links. The results indicated that the ideal transmitter beam divergence angle was approximately 15° for a 1 m transmission distance, regardless of water turbidity [158].

8.4. Magnetic induction communication

Over the past decade, underwater magnetic induction communication has attracted considerable attention as an emerging method for underwater communication. The advantages of magnetic induction communication include its underwater channel characteristics such as minimal multipath interference, reduced Doppler effect, and cross-media transmission capability. Harbin Engineering University proposed a 2D transceiver architecture for full-duplex communication that exploited the directionality of the magnetic field. The study concluded with a discussion highlighting potential future research directions [159]. Xidian University (China) introduced a bidirectional full-duplex magnetic induction communication (BFMC) based on backscattering, which enhances the energy efficiency of magnetic induction communication. Simulation results demonstrate that BFMC, combined with the proposed joint beam-forming/timing scheme, can effectively reduce the average energy consumption and improve energy efficiency [160]. The China University of Mining and Technology proposed an energy-efficient submarine multiple-input–multiple-output (MIMO) acoustic cooperative wireless sensor network that includes a cooperative MIMO formation and relay link setup. Simulation results show that this method improves data collection efficiency, reduces energy consumption at the main nodes, and increases coverage and network extension [161]. The advantages and disadvantages of the underwater communication methods are listed in Table 8.

AUVs transmit information during underwater operations to provide feedback on their operational status and transfer the collected environmental data via high-speed communications. This approach enhances the operational efficiency by reducing the need for AUVs to return to the surface. Therefore, a key research direction in underwater AUV communication technology is to achieve high-speed, accurate, and cross-media communication. China has achieved significant developments in underwater communication, such as conducting a live broadcast from the Fendouzhe HOV at a depth of 10 000 m using acoustic communication and achieving an

Table 8
Advantages and disadvantages of underwater communication methods.

Method	Application scenario	Advantage	Disadvantage
Electromagnetic	Shallow water	Low latency Capable of transmission in both water and air	Large antenna Short range in water
Acoustic	Deep water	Ultra-long-range underwater transmission	High latency High bit error rate Multipath effect in shallow water
Optical	Clear water	Low latency High-speed data transfer	Affected by water quality High angle requirements
Magnetic induction	All sea areas	Low latency High data-transfer rate Simple antenna structure Capable of transmission in both water and air	Significant signal interference Long-distance communication is not feasible

underwater acoustic communication range of 30 km. This technology has reached state-of-the-art levels in the global community.

9. Artificial intelligence in AUVs

With the development of AI technology, missions that were previously considered difficult can now be completed. As AUVs are required to perform increasingly complex underwater missions, integrating AI technology can effectively improve their intelligence and enhance their underwater operational capabilities. Currently, AI is applied to AUVs for target recognition, obstacle avoidance, and intelligent decision making.

9.1. Target recognition

The accuracy of underwater AUV target recognition is critical for detection, rescue, and safety. Target recognition technology allows an AUV to accurately identify underwater targets or obstacles, which is essential for environmental awareness, operational missions, and obstacle avoidance. With the rapid advancement in of AI technology, applying AI to target recognition in AUVs can significantly enhance recognition accuracy.

The Ocean University of China proposed an autonomous target recognition system for AUVs based on the SSS. This system used an attention mechanism and integrated learning to meet the requirements of underwater target recognition accuracy and real-time performance. The system was tested in sea trials using the Sailfish-324 AUV (Ocean University of China) to validate the algorithm [162]. Researchers at the Shenyang Institute of Automation, Chinese Academy of Sciences developed a deep-learning-based target recognition algorithm for SSS images. This algorithm employs the You Only Look Once (YOLO) version 7 network with multiscale information fusion and attention mechanisms to detect targets in sonar images. The effectiveness of the algorithm was validated in lake trials using the TS-100 AUV (Shenyang Institute of Automation, Chinese Academy of Sciences) [163]. The North Atlantic Treaty Organization Science and Technology Organization proposed an autonomous target recognition method based on the FLS as a naval mine countermeasure. This method combines a detector and classifier using a convolutional neural network (CNN) model with a probability grid graph. The effectiveness of the algorithm was confirmed using anti-mine data collected by the BIONDo AUV (Iqua Robotics, Spain) [164]. The Naval University of Engineering proposed the

detection transformer-YOLO target recognition model for rapid detection and the blended hybrid dilated convolution and pyramid split attention UNet object segmentation model for high-precision segmentation. The effectiveness of the algorithm was verified through ship wrapping and mine targeting experiments using the Black Shark I-A AUV (Deepinfar Ocean Technology Inc., China) [165]. The University of Florence introduced a CNN-based autonomous recognition method for underwater targets using the FLS. The effectiveness of this algorithm was validated through port experiments using the FeelHippo AUV [166]. Incorporating AI technology into AUV target recognition can significantly improve accuracy. However, owing to the high computational cost and need for pretraining in AI technology, running this technology on the low-performance computers of AUVs is challenging. Therefore, the development of efficient, highly accurate, and computationally lightweight target recognition methods is an important research direction for target recognition in AUVs. China has made significant breakthroughs in AI-based target recognition by successfully identifying sand dunes, sunken ships, mines, and other targets using SSS.

9.2. Obstacle avoidance

As AUVs undertake increasingly complex underwater missions, including those operating in unknown environments, effective obstacle avoidance after completing target recognition is vital to ensure autonomy and safety during underwater operations. The problem of avoiding dynamic obstacles can be addressed by applying AI technology to process high-dimensional information and handle nonlinear missions, thereby improving the obstacle-avoidance efficiency of AUVs.

Researchers at the Southeast University (China) proposed an obstacle detection method based on the FLS and YOLO version 3 algorithms along with an AUV obstacle-avoidance strategy based on deep reinforcement learning (DRL). Efficient AUV obstacle-avoidance paths were planned and verified through simulation experiments [167]. Harbin Engineering University is researching obstacle-avoidance problems for AUVs using DRL in unknown environments. An event-triggered soft actor-critic AUV collision-avoidance algorithm was proposed. The feasibility and effectiveness of the proposed algorithm for safe collision avoidance in AUVs were verified in a simulation environment [168]. Jiangsu Ocean University (China) designed an obstacle-collision prediction model by combining sequence sample data with long short-term memory networks to enhance the learning capability of the sequence data. The effectiveness of the algorithm was verified through 3D path-planning simulations [169]. Yanshan University (China) proposed a deep learning method called the deep deterministic policy gradient to minimize the cost function and plan collision-free paths for AUVs. Blue ROV2 (Blue Robotics Inc., USA) was used to experimentally verify the effectiveness of the obstacle-avoidance method [170].

With the application of AI technology for target recognition, obstacle avoidance based on identified obstacles can significantly improve the safety of AUVs. However, similar to target recognition, the use of AI in obstacle avoidance is limited by high computational costs and the need for pretraining. Therefore, lightweight obstacle-avoidance methods must be further explored. China holds a leading position in the field of AI for AUV obstacle avoidance and has conducted substantial research in this area. However, most current research on AI in the field of obstacle avoidance is still in the simulation stage, and its transition to experimental or practical applications must be accelerated.

9.3. Intelligent decision making

As AUVs continue to evolve intelligently, they are increasingly assigned to more complex and challenging underwater missions,

which place higher demands on their decision-making systems. For example, after completing missions, such as target recognition and obstacle avoidance, AUVs must autonomously make decisions regarding path planning and mission allocation. In this context, autonomous decision making is essential for completing underwater missions. AI technologies can continuously optimize mission execution in complex environments, thereby enhancing AUV operational performance.

Tianjin University proposed a sustainable reinforcement learning method to achieve robust path planning for AUVs under time-varying ocean-current disturbances. The effectiveness of the algorithm was validated through simulations [171]. Anhui University (China) introduced a DRL-based path-planning method for AUVs in 3D unknown environments. Simulation results demonstrated that this method improves AUV path planning in such environments, thereby accelerating training convergence and enhancing AUV attitude stability [172]. Babol Noshirvani University of Technology (Islamic Republic of Iran) proposed adaptive motion-planning and obstacle-avoidance techniques for AUVs based on DRL. The simulation results show that the motion-planning system can accurately guide the AUV with six degrees of freedom toward the target [173]. Florida Atlantic University (USA) developed a framework that integrates path planning and tracking control using strategy-gradient DRL to address path-planning and tracking issues. The effectiveness of the algorithm was verified through simulation experiments [174]. The applications of the aforementioned AI technologies in AUVs are summarized in Table 9.

Introducing AI technology into the intelligent decision-making processes of AUVs can effectively improve their operational efficiency in complex environments and reduce the time spent on missions. However, because of the complexity of underwater environments and presence of multiple unknown factors, a high demand has been exhibited for real-time perception and target recognition. Therefore, the development of robust intelligent decision-making technology is a major research direction in AUV development. In current research on intelligent decision-making systems in AUVs, although China has conducted several studies, the focus is primarily on theoretical research or simulations. Consequently, considerable work remains before intelligent decision-making technologies can be applied to AUVs.

10. Application and analysis of AUVs

The applications of AUVs can be classified into military and civilian applications based on the tasks that they perform underwater. Depending on the number of AUVs used in a specific scenario, they can be divided into single and multiple cooperative AUVs. The applications of AUVs in each category are subsequently introduced and analyzed.

10.1. Military applications

AUVs possess a high degree of autonomy underwater, which allows them to conduct long-term underwater operations while carrying various sensors for environmental monitoring. They can also be launched from submarine torpedo tubes, making them highly valuable for military applications. Currently, they are the most widely used underwater vehicles in military applications. The HUGIN 1000 AUV features high-accuracy seabed measurements and positioning capabilities. From 2001 to 2002, it was involved in military missions, including surveying the submarine US Ship Nautilus near the North Pole [175]. The REMUS 100 AUV has also participated in numerous military applications, including its first deployment in the northern Persian Gulf in 2003 for mine-countermeasure operations. During the Ice Exercise 2020, MIT collaborated with General Dynamics Corporation (USA) to deploy the Bluefin-21 Macrura AUV under six feet of ice, demonstrating new navigation technologies [176]. During the robotic experimentation and prototyping with maritime unmanned systems exercise in Portuguese waters, the Gavia AUV was used for beyond-line-of-sight intelligence, surveillance, and reconnaissance, supporting mine-countermeasure operations aimed at facilitating the remote monitoring of amphibious landing areas.

10.2. Civilian applications

AUVs have significant applications in the civilian world and exhibit strong underwater mapping and detection capabilities; consequently, they are extensively used in applications such as aircraft wreckage searches, underwater pipeline inspections, seabed bathymetry mapping, and marine fauna localization and tracking. In 2009, the REMUS 6000 AUV was involved in the search for Air France Flight 447, successfully locating the wreckage and crucial black boxes. Bluefin-21 is a highly modular AUV that is capable of carrying various sensors and payloads. In 2014, it assisted in the search for Malaysia Airlines Flight MH370. The University of Gothenburg in Sweden used the HUGIN AUV to extensively explore the Dotson Ice Shelf in West Antarctica. After 27 days of continuous work, they successfully mapped ice formations on the underside of the shelf [177]. WHOI uses the Mesobot AUV to explore the “twilight zone” in oceans. The AUV can automatically track free-floating hydromedusae and larval marine animals at depths of 200 m in Monterey Bay, USA. In addition to these tracking missions, vehicles can perform preprogrammed missions to collect imagery and sensor data [178]. The University of Pisa uses AUVs to detect underwater gas emissions. It utilized FLS acoustic images to autonomously identify gas leaks in real time and effectively detected artificially reproduced underwater gas leaks using the FeelHippo AUV [179]. The University of the Highlands and Islands (UK) uses AUVs for research on underwater elasmobranch ecology,

Table 9
Application of the AI technology in AUVs.

Application	AI method	Purpose
Target recognition	MobileNetV3 CNN YOLO version 7	Accurately identifying targets or obstacles to ensure the successful completion of underwater missions of AUVs
Obstacle avoidance	DRL Long short-term memory networks Deep learning	Accurately avoiding obstacles to ensure the safety of the AUV and efficient completion of missions
Intelligent decision making	Sustainable reinforcement learning DRL	Complete and accurate path planning, along with reasonable mission allocation, to improve the efficiency of the AUV

employing the Gavia AUV with a camera to observe marine life [180].

10.3. Intervention applications

Similar to ROVs, certain AUVs are equipped with mechanical devices such as robotic arms to perform underwater missions. These AUVs are referred to as intervention AUVs. They can perform tasks, such as the maintenance of marine equipment and underwater welding. The Girona 500 I-AUV, developed by the University of Girona, features a lightweight four degrees-of-freedom electric manipulator, video system, and associated control electronics. The AUV is designed to conduct a two-stage autonomous underwater intervention mission. First, an initial video survey phase is used to locate a specific object, followed by the retrieval of this object using a hook attached to the robotic arm. The University of Technology Sydney (Australia) employs submerged pile inspection robot (SPIR3.0, Centre for Autonomous Systems of University of Technology Sydney, Australia) as an intervention AUV, which utilizes sliding mode impedance control (SMIC) for full-contact intervention, transitioning from a noncontact to a contact phase. Their experimental results demonstrated the effectiveness of SMIC in facilitating contact interventions using AUVs [181]. A hydropathic dual-arm intervention AUV called Cuttlefish, developed by the German Research Centre for Artificial Intelligence, can adopt various poses in a water column to reach difficult-to-access objects using two specially designed manipulators [182]. IFREMER proposed a comprehensive control architecture for docking a VORTEX intervention AUV (IFREMER) to a reception structure, which use sonar and video image processing along with navigation data from conventional sensors [183].

10.4. Cluster applications

The aforementioned AUV applications are conducted using individual AUVs. However, owing to the operational capacity limitations of single AUVs, their efficiency and capabilities are restricted. To enhance the operational capacity within the same timeframe, AUVs are now deployed in clusters to perform underwater tasks, significantly improving the efficiency and stability of AUV-based detection. The European Union (EU) initiated the GREX project from 2006 to 2009. By 2009, marine tests verified that multi-AUV systems can achieve cooperative navigation and control through acoustic communication [184]. The TS-100 AUV is designed to meet high spatiotemporal resolution and 3D network observation needs for various factors. Through underwater network communication, multiple explorations of 100 AUVs demonstrated autonomous formations and cooperative search capabilities. The Kiel University of Applied Sciences (Germany) utilized Hansel and Gretel AUVs (University of Applied Sciences Kiel, Germany) to form heterogeneous multi-AUV clusters, maintained communication through acoustic, inductive, and optical links, and validated the effectiveness of clusters through experiments [185]. From 2015 to 2018, with EU financing, the “Widely Scalable Mobile Underwater Sonar Technology” (WiMUST) project witnessed the development of advanced cooperative and networked control/navigation systems to enable a group of marine robots (both at the surface and submerged) equipped with acoustic sources and towed acoustic streamers to perform geotechnical seismic surveys in a fully automatic manner. For the first time worldwide, a mission was performed in 2018 at sea in Sines, Portugal, with a fleet of seven autonomous marine robots cooperatively performing high-resolution 3D sub-bottom mapping using EvoLogics acoustic communication and positioning systems [186]. From 2014 to 2015, Harbin Engineering University conducted heterogeneous multi-AUV cooperative sea trials in Weihai,

China. This experiment utilized radio and acoustic communication systems to perform various ocean applications, including graph path and target tracking and behavior-based cooperative formations. AUVs are extensively used in the underwater environment, particularly in both civilian and military applications. The US and countries of the EU have equipped their navies with AUVs and are using them in naval exercises and real-life combat situations. They also use AUVs to search for crashed aircrafts, monitor marine life, and map bathymetry terrain and ice shelves, effectively expanding their applications. In contrast, China is still in the early stages of AUV application. AUVs are primarily used to verify specific indicators or explore specific areas. Therefore, China should intensify practical research on AUVs, capitalize on their underwater advantages, and accelerate their application process.

11. Conclusions and future research

11.1. Conclusions

AUVs are efficient and accurate marine exploration tools. Although AUVs are not yet as widely used as ROVs, their effectiveness will become more apparent as AUV technology continues to develop and mature. Based on a review of the current key technologies of AUVs, the following conclusions can be drawn:

(1) A well-conceived overall design can enhance the operational capabilities of AUVs in various scenarios. Utilizing a modular design for AUVs facilitates flexible configuration and adjustment based on different tasks, thereby ensuring robust performance across diverse operational contexts.

(2) Integrating fuel cell technology with underwater wireless charging systems can improve the endurance of underwater AUVs. Currently, Li-ion batteries are primarily used in underwater charging technology. Research on wireless charging technology for fuel cells has provided valuable insights on enhancing the long-term underwater operational capabilities of AUVs.

(3) Among the existing planning and control technologies, AI has the potential to achieve superior performance. However, computational power-related limitations can restrict the ability of AUVs to make effective decisions in dynamic environments. Enhancing the application of AI in AUVs is important.

(4) Different underwater environmental sensing technologies exhibit varying advantages and limitations; consequently, a single sensing system in an AUV is unable to fully perceive its surrounding environment and make accurate decisions. Therefore, data from various sensing systems must be integrated to improve the accuracy of environmental perception.

(5) Geophysical field navigation enables AUVs to perform long-term underwater navigation. However, limitations related to the need for prior geophysical maps have restricted its widespread application. Combining multiple geophysical field navigation methods can enhance the navigation performance in different areas, thereby providing AUVs with extended underwater navigation capabilities.

(6) Underwater communication primarily relies on acoustic methods, which have limited bandwidth and low transmission rates. Optical and magnetic induction communication technologies offer high-speed and accurate underwater communication capabilities, and even enable cross-medium communication, providing promising directions for the development of underwater communication technology. Future research will generate considerable activities related to the development of multimodal communication systems that effectively adapt the type of communication to the perceived operating conditions.

(7) With the application of AI technology, AUVs can achieve enhanced target recognition and decision-making capabilities.

However, the current research on AI technology is largely based on simulations or data playback. Therefore, the practical application of AI technology in AUVs must be strengthened and their operational capabilities in complex underwater environments must be improved.

(8) With the increasing value of oceans and their resources, AUVs exhibit significant potential for applications in both civilian and military fields. In current AUV applications, the limited operational capability of a single AUV may prevent it from completing missions in the event of a failure. Therefore, enhancing the capabilities of AUV clusters and improving their operational performance in complex environments are essential.

11.2. Future development

With the increase in human exploration of oceans, AUVs play an increasingly important role in areas such as underwater exploration, environmental monitoring, marine research, military operations, underwater construction, and search and rescue. This underscores the significant potential in the future development of AUVs. A summary of the primary directions of AUV development based on current research is presented subsequently.

(1) The design of AUVs will evolve toward a more streamlined and modular structure to enhance their maneuverability and flexibility in handling various missions. The pressure hull will undergo a structural transformation to optimize the use of internal space while maintaining pressure resistance. Materials for pressure hulls will shift toward lightweight, high-strength options, such as carbon fibers and composite materials.

(2) The energy systems of AUVs will evolve toward higher energy densities and greater safety by incorporating technologies such as LiFePO_4 batteries and fuel cells. Underwater charging technology will progress to achieve higher power output and lower losses, thereby improving the long-term operational capabilities of AUVs. The propulsion and actuator systems of AUVs will advance to achieve better maneuverability, reduced noise, and simplified control, thereby enhancing overall performance.

(3) The planning and control technologies of AUVs will evolve toward greater intelligence and cluster operations. By integrating AI technologies, such as deep and reinforcement learning techniques, into their control and planning systems, AUVs can continuously improve their autonomous decision-making capabilities in complex environments. Furthermore, AI can be applied for fault diagnosis, further enhancing its reliability.

(4) Perception technology in AUVs will advance toward multi-sensor fusion by combining the advantages of various sensors such as optical, acoustic, and biomimetic sensors. New sensing methods such as biomimetic electric field sensing will be developed to further improve the accuracy and reliability of AUVs in complex environments.

(5) Navigation and positioning technologies in AUVs will evolve toward the integration of multiple navigation methods, such as inertial and geophysical navigation techniques, to enhance the performance in complex environments by fusing data from multiple sources. Moreover, new navigation methods that combine AI with biomimetic navigation will be explored to improve the navigation performance during long-distance missions.

(6) The communication technology of AUVs will advance toward the integration of multiple communication methods, leveraging the advantages of each to achieve long-range, high-speed, and low-latency communication. New communication methods, such as quantum communication, will be developed to overcome the limitations of underwater communication and enable high-speed data transmission. Cross-media communication technologies will also be developed to satisfy the demands of complex underwater missions.

(7) In the future, AI technology will be more widely applied in AUVs, driving them toward higher intelligence, real-time autonomous decision-making, and multi-mission clusters. As new AI technologies such as embodied intelligence and large language models mature, their integration into AUVs will further enhance their operational capabilities in complex environments.

(8) With the growing demand for ocean exploration, AUVs are expected to have an increasing range of applications. In the future, AUVs will hold significant potential in civilian applications such as marine environment monitoring, resource exploration, and marine biological observation. They will also be extensively used for exploring extreme environments, such as polar regions and deep-sea environments. Moreover, the use of AUV clusters will enhance exploration capabilities and improve efficiency.

AUVs with functionalities and operational capabilities that are nonexistent in other underwater vehicles are essential tools for efficient and accurate ocean exploration. Therefore, increased research and investment in key AUV technologies are crucial for accelerating their development. This progress will offer significant economic and environmental benefits for exploring and understanding oceans.

CRedit authorship contribution statement

Dong Ma: Writing – original draft, Investigation. **Ye Li:** Supervision, Funding acquisition, Conceptualization. **Teng Ma:** Writing – review & editing, Funding acquisition, Conceptualization. **António M. Pascoal:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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